

Pure Pursuit vs. Stanley for DV Autocross

Lateral controller comparison in MATLAB/Simulink · Formula Student Driverless · Alex Demir

WHAT WAS BUILT

One Simulink model in which **only the steering law changes**. Track, speed reference, longitudinal PI, steering actuator (saturation, rate limit, first-order lag), 100 Hz control rate and fixed-step *ode4* solver are identical for both controllers, so every measured difference belongs to the steering law. MATLAB + Simulink only: both laws, the vehicle model, the track generator and the KPIs are written from scratch, and the model is generated by a script rather than shipped as a binary.

- Track** — closed filleted polygon, cone-delimited: 212.6 m lap, 12 m minimum turning diameter, 53.2 m longest straight, 3.5 m wide, closing to 0.23 mm. Compliant with Formula Student Rules 2026 D 8.1. Contains a long straight into a hairpin, a slalom, a decreasing-radius corner and a fast sweep.
- Controllers** — pure pursuit at the rear axle, $\delta = atan(2L \sin \alpha / L_d)$ with speed-scheduled lookahead; Stanley at the front axle, $\delta = e_{\psi} + atan(-k_e e_y / (k_{soft} + v))$. Gains chosen by grid search against one shared objective, scored outside the design envelope.
- Verification** — 21 analytical unit tests, including a closed-form check that each law returns the Ackermann angle $atan(L/R)$ on a circle. It caught a real defect: the pure pursuit goal point was marched along arc length rather than found on the chord. The Simulink model agrees with an independent RK4 reference loop to 4.4×10^{-11} m over a lap.

RESULTS — NOMINAL LAP, 3.5 M TRACK

KPI	Pure Pursuit	Stanley	better
1 lap time	17.960 s	18.180 s	PP by 0.22 s
2 max cross-track error	0.902 m	0.129 m	Stanley by 7.0×
3 RMS cross-track error	0.408 m	0.055 m	Stanley by 7.4×
4 RMS steering rate	8.73 °/s	21.12 °/s	PP by 2.4×
5 cones Down or Out	0	0	tie
FSG DV Autocross points (D 9.3.2)	100.00	98.87	PP by 1.13

Why the less accurate controller laps faster. Both receive the same speed reference at the same arc-length station, so lap time reduces to distance driven. Pure pursuit steers along a chord to a point ahead on the centreline, and a chord passes inside the arc, so it cuts every corner: 208.54 m driven against Stanley's 210.17 m on a 212.64 m centreline. Held to Stanley's accuracy it laps in 18.320 s — slower. The reference is the centreline, not a racing line, so a large tracking error is not automatically a fault: KPIs 2 and 3 are read here as **margin** metrics.

THE DECIDING EXPERIMENT — THE WIDTH THE RULES GUARANTEE

Everything above is on a 3.5 m track; **D 8.1.1 guarantees only 3 m**. Clearance is the gap between the outer edge of the car and the cone line at the worst point of the lap.

width	speed	PP clearance	PP cones	PP points	ST clearance	ST points
3.50 m	×1.00	+0.25 m	0	100.00	+1.02 m	98.87
3.50 m	×1.30	+0.11 m	0	100.00	+1.04 m	99.09
3.00 m	×1.00	-0.00 m	2	80.29	+0.77 m	100.00
3.00 m	×1.30	-0.14 m	6	50.53	+0.79 m	100.00

Verdict: Stanley. Pure pursuit's 1.13-point lead exists only on a track wider than the rules promise. At the guaranteed minimum it loses 20 to 50 points to 2 s cone penalties, while Stanley never drops below 0.77 m of clearance or touches a cone at any width or speed tested. Robustness to pose latency (0–100 ms) and position noise (0–10 cm) is equivalent once gains are chosen outside the design envelope. Stanley gives up 0.22 s a lap and buys margin for the errors this model omits — SLAM drift, a mis-detected cone, a wet patch. Pure pursuit stays the better answer where the steering actuator is the binding constraint, or where the loop delay cannot be characterised.

JUDGEMENT — NEITHER LAW BELONGS ON A HIGH-SPEED CAR

Both controllers are geometric, derived from a kinematic bicycle model with no tyre model and no slip. For an F1-class car, or any vehicle at the grip limit, that disqualifies **both**, and a dynamic-model controller such as LQR or MPC with tyre stiffness is required. Of the two, pure pursuit is the less suitable at speed despite winning on lap time here: its lookahead grows with speed, so its corner cutting grows with speed, and its whole advantage comes from deliberately running closer to the track limit. When the cost of an unmodelled slip is leaving the track rather than brushing a cone, spending margin to buy tenths is the wrong trade.

Limitations — kinematic vehicle model, so no understeer at the grip limit; the reference is the centreline rather than an optimised racing line, which is why lap time largely measures how far inside it each law runs; one track layout; gains tuned on this track; a shared and deliberately simple longitudinal controller. **Reproduce** — `addpath('src','tests');`; `run_tests`; `run_all` (MATLAB + Simulink only; the non-Simulink parts also run in GNU Octave).